Video Demonstration Script: IEM Admission System Software

Production-Ready Walkthrough Script from GitHub to Live Cloud Deployment

Target Duration:	~9 Minutes 30 Seconds (Strictly ≤ 10 Minutes at ~135 words/minute)
Initial View:	Google Chrome on Windows 11 with GitHub Repository tab open (Susnata-Maiti/IEM-Admission-System)
Live URL:	https://iem-admission-system-nexus-p4.vercel.app/ (Vercel Edge CDN + MongoDB Atlas Cloud)
Interactive Docs:	https://iem-admission-system-nexus-p4.vercel.app/docs (FastAPI OpenAPI / Swagger UI)
Team Details:	Team Number: P4 | Team Name: Nexus | Batch: 2024-2028 (Section A, 5th Sem)
Team Members:	Susnata Maiti (A-11, Frontend) | Sreejita Dey (A-29, Backend) | Swapnil Ghosh (A-04, Database) | Debadrita Dutta (A-12, Testing)
Faculty In-Charge:	Prof. Sangeeta Sarkar & Prof. Saikat Dutta
Project Sponsor:	Dr. Satyajit Chakrabarti (Director, Institute of Engineering & Management, Kolkata)

1. Pre-Recording Setup & Environment Checklist

Screen Setup:	Windows 11 Desktop display set to 1920×1080 (1080p). Hide browser bookmarks bar (Ctrl+Shift+B).
Browser Tabs:	Tab 1: GitHub repo at top. Tab 2: FastAPI Swagger docs (/docs) ready in background.
Sample Asset:	Valid sample PDF marksheet under 5MB (e.g. marksheet_sample.pdf, ~1.2MB) accessible on Desktop.
Audio & Pacing:	Quiet room with headset/mic. Target speed: ~130-140 wpm. Allow 1-2s pauses for UI transitions.

2. Timing & Scene Overview Table

The video demonstration is organized into 8 modular scenes totaling 9m 45s, staying strictly below the 10-minute limit:

Timestamp	Scene / Section	Primary Visual Focus	Topics Covered
0:00 - 0:45	**Scene 1: Launch & Intro**	GitHub Repo README in Chrome	Team intro, PCCCS504 context, 20-day timeline, clicking live link.
0:45 - 2:00	**Scene 2: Landing & Reg**	Live Web Portal (/) Onboarding	IEM crest, Google Sans font, @gmail.com validation, PBKDF2 hashing.
2:00 - 4:15	**Scene 3: 4-Step Wizard**	Application Stepper (/application)	Personal data, academic scores, B.Tech CSE, PDF upload (<5MB), form lock.
4:15 - 5:30	**Scene 4: Student Dashboard**	Student Dashboard (/dashboard)	Application ID, tracking timeline, marksheet preview, immutable lock.
5:30 - 7:15	**Scene 5: Admin Workspace**	Admin Console (/admin)	Real-time KPI metrics, <1s search filter, Table/Card views, state updates.
7:15 - 8:15	**Scene 6: Director View**	Director Dashboard (/director)	Executive bird's-eye metrics, read-only mode, one-click CSV export.
8:15 - 9:15	**Scene 7: Architecture & QA**	Swagger Docs (/docs) / Diagrams	FastAPI ASGI engine, Dual-Engine DB abstraction, 100% basis path tests.
9:15 - 9:45	**Scene 8: Conclusion**	GitHub Repo / Presentation Slide	Deliverables summary, faculty acknowledgments, and closing thanks.

3. Detailed Two-Column Production Script (Visual Actions & Voiceover)

Follow the left column for on-screen actions and read the right column word-for-word:

Scene 1: Introduction & Launch from GitHub • 0:00 - 0:45 (45s)

Visual Cue & On-Screen Action	Spoken Narration (Voiceover Script)
Screen: Chrome browser showing GitHub repository *Susnata-Maiti/IEM-Admission-System*. **Action:** Cursor hovers over repository title and team header in README.md. Scroll smoothly down to Section '■ *Live Production Deployment*'. **Action:** Hover mouse cursor over the live Vercel URL and click https://iem-admission-system-nexus-p4.vercel.app/.	*"Hello everyone. Welcome to the technical demonstration of the **IEM Admission System Software**, engineered for the Software Engineering and Project Management Lab (PCCCS504) at the Institute of Engineering & Management, Kolkata.* *I am **Susnata Maiti**, Frontend Developer for **Team Nexus (P4)**. I am presenting on behalf of our team comprising **Sreejita Dey** as Backend Developer, **Swapnil Ghosh** as Database Lead, and **Debadrita Dutta** as Test Lead, under the guidance of our faculty in-charge **Prof. Sangeeta Sarkar**, **Prof. Saikat Dutta**, and our Project Sponsor and Director, **Dr. Satyajit Chakrabarti**.* *Our objective was to deliver a resilient, user-focused college admission platform within an accelerated 20-day timeline. As documented in our repository README, the application is deployed live on Vercel with a global CDN edge, backed by MongoDB Atlas Cloud. Let us click our live production link and walk through the complete admission lifecycle in real time."*

Scene 2: Landing Page, Design Standards & Student Registration • 0:45 - 2:00 (75s)

Visual Cue & On-Screen Action	Spoken Narration (Voiceover Script)
Screen: Browser opens *https://iem-admission-system-nexus-p4.vercel.app/*. **Action:** Point out the official IEM crest and clean Google Sans font. Toggle to 'Register' tab. **Action:** In Name, type 'Susnata Maiti'. In Email, enter an invalid address 'susnata@yahoo.com' and click Register to demonstrate inline @gmail.com domain error toast. **Action:** Correct email to 'maitisusnata135531@gmail.com'. In Password, type 'Student@2026!'. Click 'Create Account / Register'. Show success toast and automatic login.	*"Here on our live production portal, the user interface adheres strictly to our SRS design specifications: featuring the official IEM laurel wreath crest, responsive container layouts, and clean typography rendered in **Google Sans**.* *The system implements strict Role-Based Access Control separating Student Applicants, Admissions Administrators, and the Executive Director. Let us begin by registering a new student applicant.* *Notice our client-side validation guard: if I input a non-Gmail address, the portal immediately rejects it because student registrations are restricted strictly to **@gmail.com** addresses. Once I provide my valid email—maitisusnata135531@gmail.com—and enter a password satisfying our complexity policy (minimum 8 characters with uppercase and special symbols), the client sends an asynchronous REST call to our FastAPI backend.* *On the server side, passwords are never stored in plaintext—they are securely hashed using salted **PBKDF2-HMAC-SHA256**. The backend issues a signed JWT token, logging us in seamlessly."*

Scene 3: The 4-Step Application Wizard & PDF Upload • 2:00 - 4:15 (135s)

Visual Cue & On-Screen Action	Spoken Narration (Voiceover Script)
Screen: Application Wizard (/application). **Action 1 (Step 1):** Point out 4-step progress stepper. Fill DOB: 15/08/2005, Phone: +91 98765 43210, Address: Salt Lake Sector V, Kolkata. Click Next. **Action 2 (Step 2):** Enter Class 10 marks (92.5%, CBSE), Class 12 marks (90.0%, WBCHSE), Passing Year (2024). Select Course: 'B.Tech Computer Science & Engineering'. Click Next. **Action 3 (Step 3):** Drag and drop sample PDF marksheet (~1.2MB). Show file preview card with size verification. Click Next. **Action 4 (Step 4):** Review consolidated summary. Point out Business Rule BR-02 immutability warning. Click 'Confirm & Final Submit'.	*"We are now guided into the **4-Step Application Wizard**. The top visual stepper anchors candidate progress, minimizing cognitive fatigue.* *In Step 1, personal details are sanitized and checked for mandatory fields before enabling navigation to Step 2.* *In Step 2, the candidate inputs academic scores and selects their target degree from a predefined list. Per Business Rule **BR-04**, percentages must fall strictly between 0.0% and 100.0%, enforced via Pydantic schema validation.* *Step 3 implements **Module 1: Document Upload**. In compliance with requirement **FR-05**, uploads are restricted strictly to PDF format with an enforced 5 megabyte ceiling. Both client-side JavaScript and the FastAPI multipart stream validator inspect MIME headers to protect against malicious file uploads. The file is stored safely and linked to our record.* *In Step 4, the student reviews their full application. Notice the explicit notice for Business Rule **BR-02**: once submitted, all personal, academic, and document fields become permanently immutable to protect admission integrity. Let us click 'Confirm and Final Submit'."*

Scene 4: Student Dashboard & Real-Time Tracking Timeline • 4:15 - 5:30 (75s)

Visual Cue & On-Screen Action	Spoken Narration (Voiceover Script)
Screen: Student Dashboard (/dashboard). **Action:** Highlight the generated Application ID (e.g. IEM-2026-0847) and green status badge: 'SUBMITTED'. **Action:** Point out chronological progress timeline: Application Submitted (Completed) → Under Review (Pending) → Final Decision (Pending). **Action:** Click 'View / Download Uploaded Marksheet' button to preview the PDF in a new browser tab. **Action:** Click 'Sign Out' in top navbar to return to login page.	*"The portal redirects immediately to the **Student Dashboard**, satisfying requirements **FR-10** and **FR-11**. Notice our assigned application identifier—**IEM-2026-0847**—and the active status badge marked 'Submitted'.* *Below the summary card is our chronological tracking timeline. The candidate can clearly track their progress through document triage and departmental review. Candidates can also preview or download their uploaded PDF marksheets anytime to verify their records. Per requirement **FR-09**, editing controls have been completely removed—the record is securely locked against further edits.* *Now, let us log out of the student session and log in as an Admissions Administrator to process this application."*

Scene 5: Administrator Workspace & State Machine Workflow • 5:30 - 7:15 (105s)

Visual Cue & On-Screen Action	Spoken Narration (Voiceover Script)
Screen: Login Card. Enter admin credentials: *admin@iem.edu.in* / *Admin@2026!*. Click 'Sign In as Staff / Admin'. **Action:** Arrive at /admin. Highlight top 4 metric counters: Total, Pending, Accepted, Rejected. **Action:** Point out Dual View toggle. Switch between 'Table View' and 'Card View' (NFR-03). **Action:** Type 'Susnata' in live search bar → show table filtering in under 1 second. **Action:** Click 'Review / Update Status' on Susnata Maiti's row. In status dropdown, select 'Under Review'. Enter remark: 'Academic credentials verified. Forwarded to CSE Departmental Board.' Click Update.	*"We now authenticate using the pre-seeded administrator credentials: admin@iem.edu.in. The backend verifies our role claim and opens the **Administrator Workspace**.* *At the top, dynamic metric counters aggregate application volumes in real time. To satisfy Non-Functional Requirement **NFR-03**, administrators can toggle smoothly between a **Compact Table View** for fast screening and a **Detailed Card View** for comprehensive candidate inspection.* *Notice our live search filter: as I type 'Susnata', the applications table updates in under one second without page reloading.* *Clicking 'Review' opens our candidate evaluation modal. Here, our **Finite-State Machine (Module 3)** regulates status transitions. An application in 'Submitted' state can transition to 'Under Review', or directly to 'Rejected'—provided explanatory remarks are supplied to prevent arbitrary rejections. Let us place this candidate **'Under Review'** with an official remark. The update is persisted immediately, and an immutable audit log entry is recorded with timestamp and reviewer identity per **FR-13**."*

Scene 6: Executive Director Dashboard & CSV Export • 7:15 - 8:15 (60s)

Visual Cue & On-Screen Action	Spoken Narration (Voiceover Script)
Screen: Log out of Admin. Enter Director credentials: *director@iem.edu.in* / *Director@2026!*. Click 'Sign In'. **Action:** Show Director Dashboard (/director). Point out high-level KPI cards and that status update buttons are hidden/read-only. **Action:** Click 'Export Full Applications List (CSV)'. Show browser downloading iem_applications_export.csv.	*"Next, let us demonstrate executive governance by logging in as our Director, **Dr. Satyajit Chakrabarti**.* *The **Director Dashboard (FR-20)** provides an executive overview of enrollment metrics. To enforce role separation, the Director view is strictly read-only: status mutation controls are hidden, eliminating the risk of accidental alterations during executive review.* *To satisfy requirement **FR-12**, leadership can export the entire dataset with one click. As you see, the browser immediately downloads a formatted CSV file containing complete applicant bio-data, academic scores, and lifecycle statuses for offline reporting and university accreditation."*

Scene 7: Architecture, Dual-Engine Persistence & Testing Rigor • 8:15 - 9:15 (60s)

Visual Cue & On-Screen Action	Spoken Narration (Voiceover Script)
Screen: Switch to browser tab showing FastAPI Interactive OpenAPI Docs (/docs) or System Architecture Diagram. **Action:** Scroll through REST endpoints (/api/auth, /api/student/application, /api/admin/metrics). **Action:** Highlight Dual-Engine Database Abstraction (MongoDB Atlas Cloud + SQLite3 local failover) and 16/16 tests passing.	*"Behind this user interface is a high-performance **FastAPI ASGI backend**, fully documented with interactive OpenAPI endpoints and typed Pydantic models.* *Our primary engineering breakthrough is the **Dual-Engine Database Abstraction Layer**. In production, the system connects to MongoDB Atlas Cloud. However, within academic institutions, campus firewalls frequently block Windows DNS queries for SRV records. To prevent connection drops, our database layer automatically injects public DNS resolvers (8.8.8.8 and 1.1.1.1). Furthermore, if cloud latency exceeds 4000 milliseconds, the engine silently falls back to an embedded **SQLite3** database (iem_admission.db) with zero downtime.* *Finally, our Test Lead, **Debadrita Dutta**, conducted McCabe's Cyclomatic Complexity basis path testing on our transition logic (**V(G)=10**), verifying all 10 independent paths across our automated test runner with a **100% pass rate** across both MongoDB and SQLite."*

Scene 8: Conclusion & Acknowledgments • 9:15 - 9:45 (30s)

Visual Cue & On-Screen Action	Spoken Narration (Voiceover Script)
Screen: Return to the GitHub repository tab in Chrome or the presentation cover slide. **Action:** Display Team Nexus member details and live project links. Cursor rests on the live production URL.	*"In conclusion, the **IEM Admission System** successfully fulfills all 20 Functional Requirements and 5 Non-Functional Requirements, delivered on schedule within our strict 20-day milestone plan.* *On behalf of **Team Nexus**—Susnata Maiti, Sreejita Dey, Swapnil Ghosh, and Debadrita Dutta—we express our gratitude to **Prof. Sangeeta Sarkar**, **Prof. Saikat Dutta**, and our Director, **Dr. Satyajit Chakrabarti**, for their guidance.* *Thank you very much for watching our demonstration!"*

4. Demo Accounts & Credentials Quick Reference

User Role	Login Email Address	Password	Primary Workflow to Demonstrate
New Student	maitisusnata135531@gmail.com	**Student@2026!**	Self-registration, 4-step wizard, PDF marksheet upload (<5MB), form locking.
Existing Student	priya.sharma@example.com	**Student@2026!**	Instant view of previously submitted application, immutable locking verification.
Administrator	admin@iem.edu.in	**Admin@2026!**	Review applications queue, search/filter, Table/Card views, status transitions with remarks.
Executive Director	director@iem.edu.in	**Director@2026!**	Bird's-eye metrics dashboard, read-only mode verification, one-click CSV export.

5. Presenter Delivery Guidelines

Speech Rate:	Maintain ~130 to 140 words per minute. Do not rush technical terms like PBKDF2 or McCabe Complexity.
Mouse Discipline:	Avoid erratic or circular motion. Rest cursor deliberately on the element being described.
Natural Pauses:	Allow 1-2 seconds of natural silence while views transition so the audience can see the DOM update.
Clean Desktop:	Silence Windows 11 notifications ('Do Not Disturb'). Hide bookmarks bar (Ctrl+Shift+B).